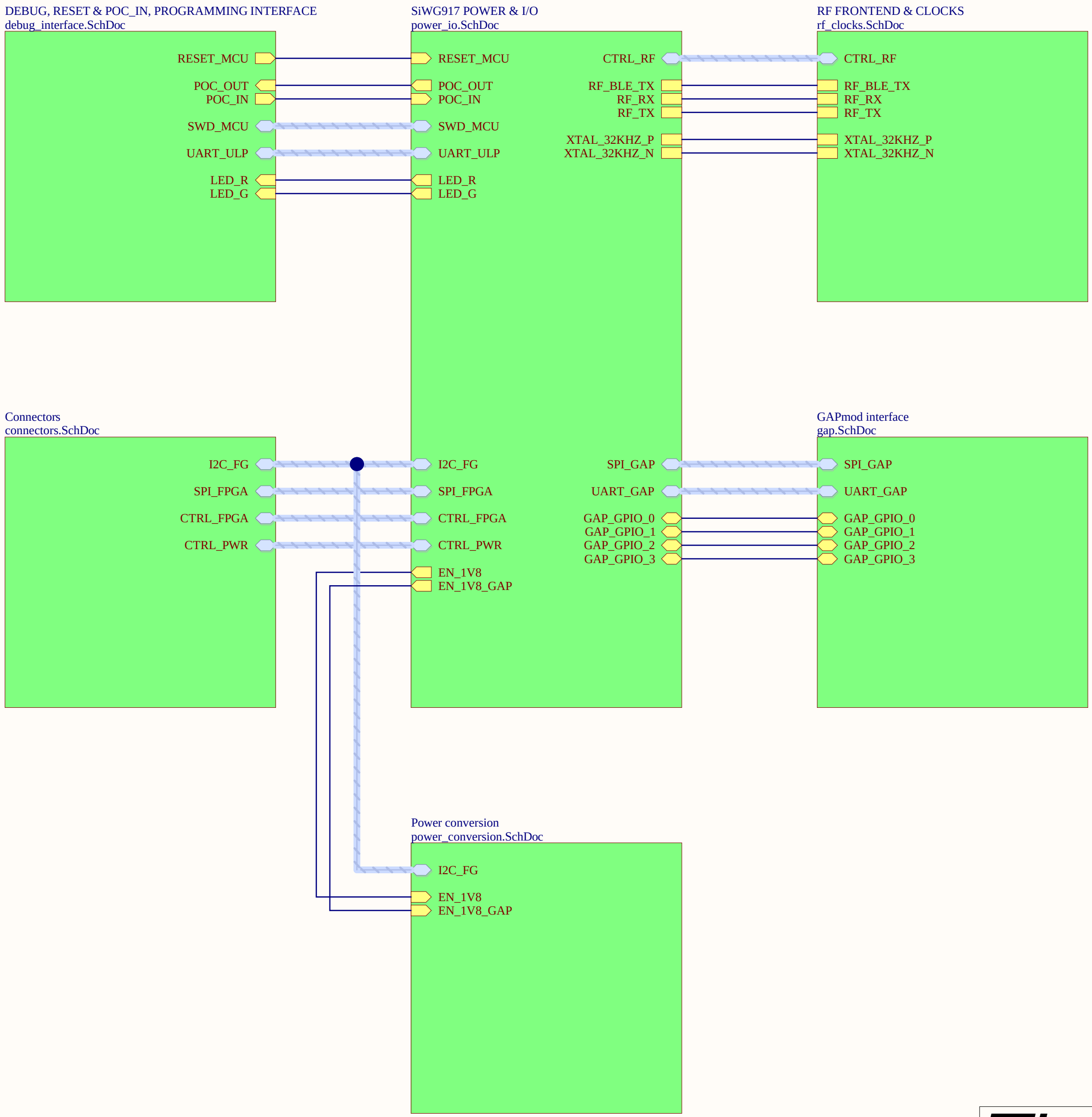

ETH
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Project: WiUS

main

High-Throughput, Low-Power WiFi for Wearable Ultrasound Systems

Laboratory: Integrated Systems Laboratory

Date: 23.04.2024 17:23:44

Rev: v1.0

Variation: all_1v8

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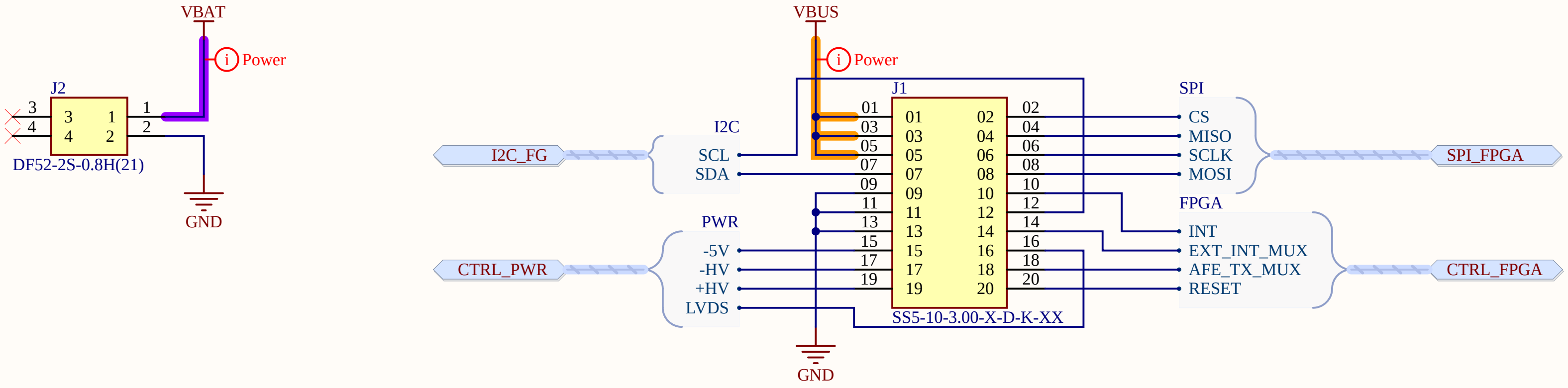

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connectors

High-Throughput, Low-Power WiFi for Wearable Ultrasound Systems

Laboratory: Integrated Systems Laboratory
Date: 23.04.2024 17:23:44
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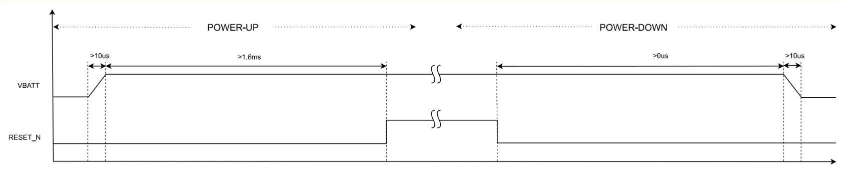
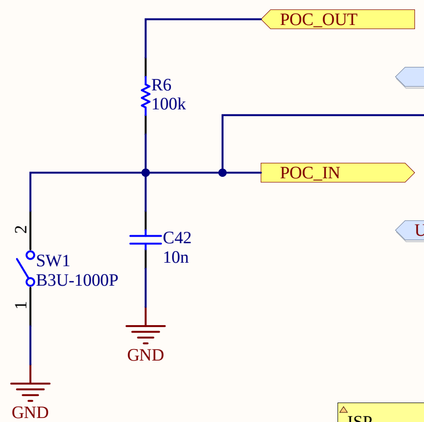
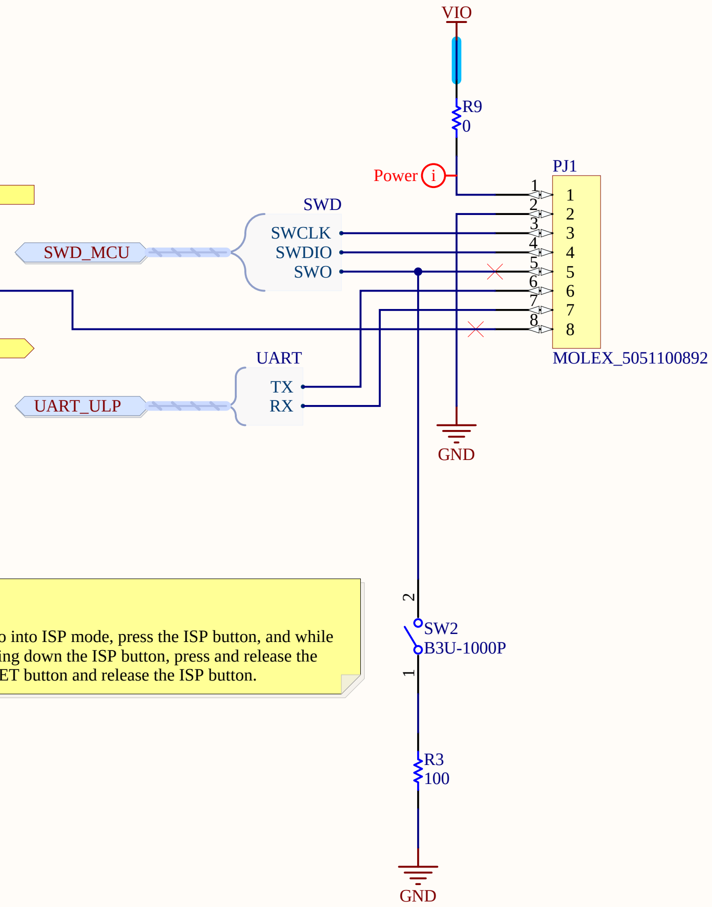
SOURCE: https://www.silabs.com/documents/public/data-sheets/siwg917-datasheet.pdf PAGE 102

The IC generates a POC (Power On Control) signal - POC_OUT that is distributed to all I/O cells to prevent the I/O cells from powering up in an undesired configuration and is also used inside the IC to safe state the IC till a valid supply is available for proper operation. This power management is functional in both power up and power down sequences.
During power up, until the UULP_VBATT_1 and UULP_VBATT_2 (VBATT supply) reach 1.6V, the POC_OUT signal stays low. Once the VBATT supply exceeds 1.6V, the POC_OUT becomes high and normal operation of the IC starts.
Once the POC_OUT becomes high, it stays high. But if VBATT becomes lower than the Blackout threshold voltage, POC_OUT becomes low.
The following figure illustrates the power up sequence when POC_OUT is connected to POC_IN. As shown in the figure below, the RESET_N is high at least 1.6ms after VBATT supply is stable. The RESET_N signal can be controlled via options like an R/C circuit or another MCU's GPIO.

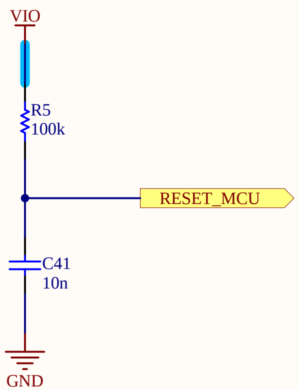


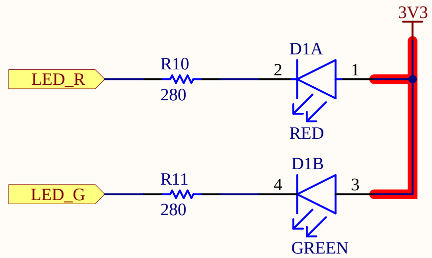




ISP

To go into ISP mode, press the ISP button, and while holding down the ISP button, press and release the RESET button and release the ISP button.





Vdd = 3.3V
If = 5mA for both
Vf,red = 1.84V
Vf,green = 1.96V

--> R,red = 292Ohm
R,green = 268Ohm

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Project: WiUS
Debug interface
High-Throughput, Low-Power WiFi for Wearable Ultrasound Systems

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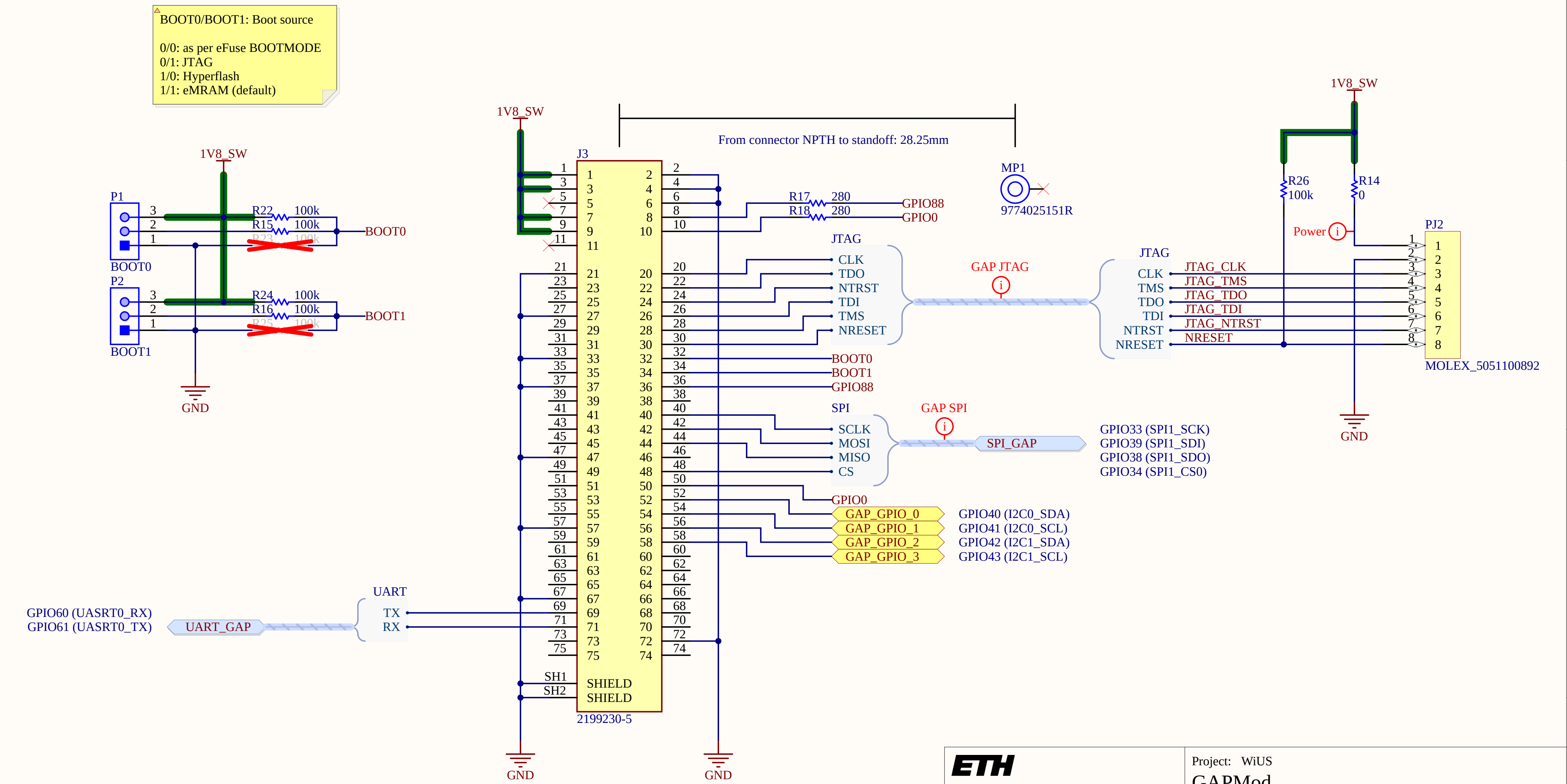
File:



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Project: WiUS

GAPMod

High-Throughput, Low-Power WiFi for Wearable Ultrasound Systems

Laboratory: Integrated Systems Laboratory
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Rev: v1.0
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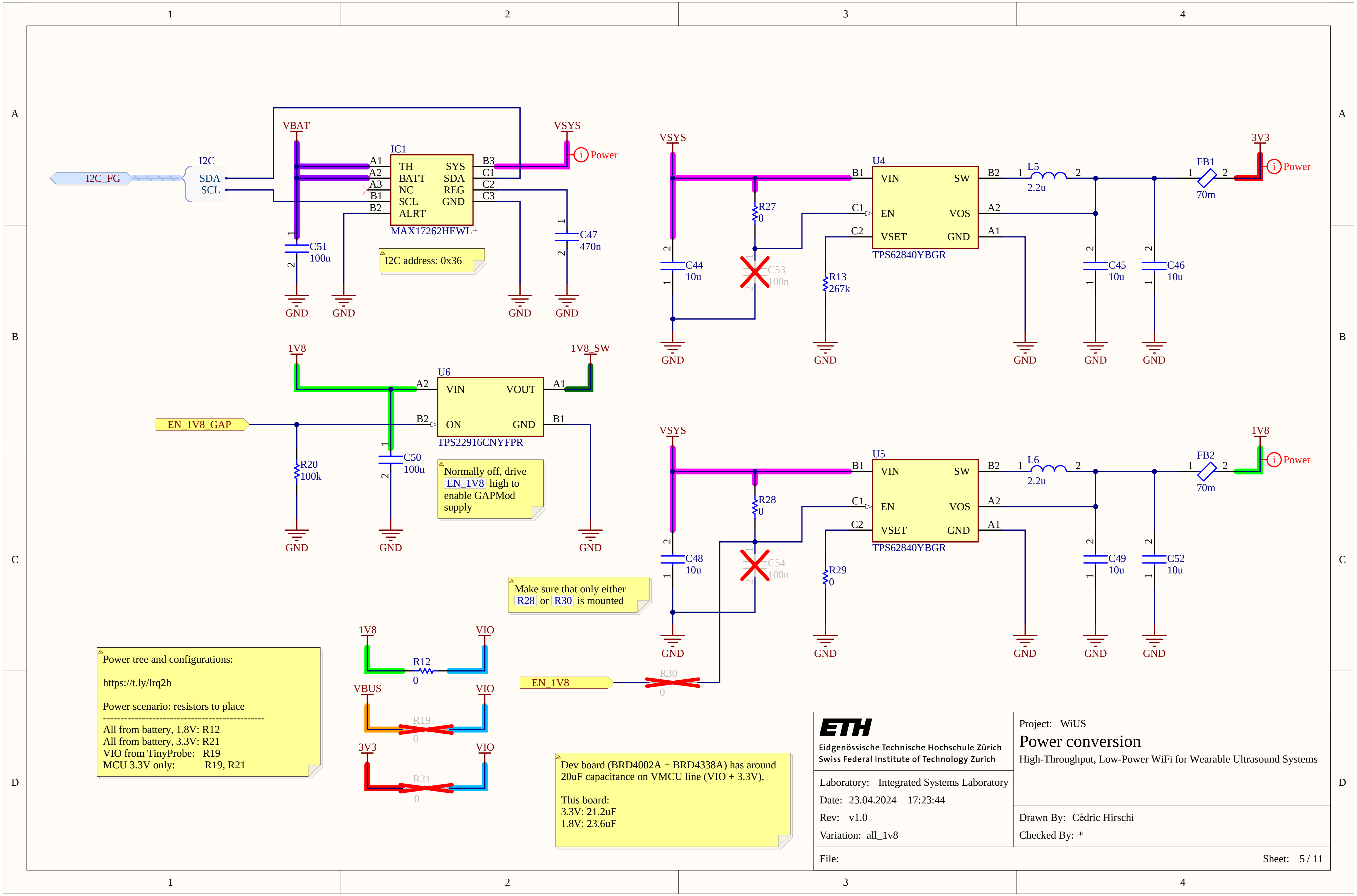

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Project: WiUS

Power conversion

High-Throughput, Low-Power WiFi for Wearable Ultrasound Systems

Laboratory: Integrated Systems Laboratory
Date: 23.04.2024 17:23:44
Rev: v1.0
Variation: all_1v8

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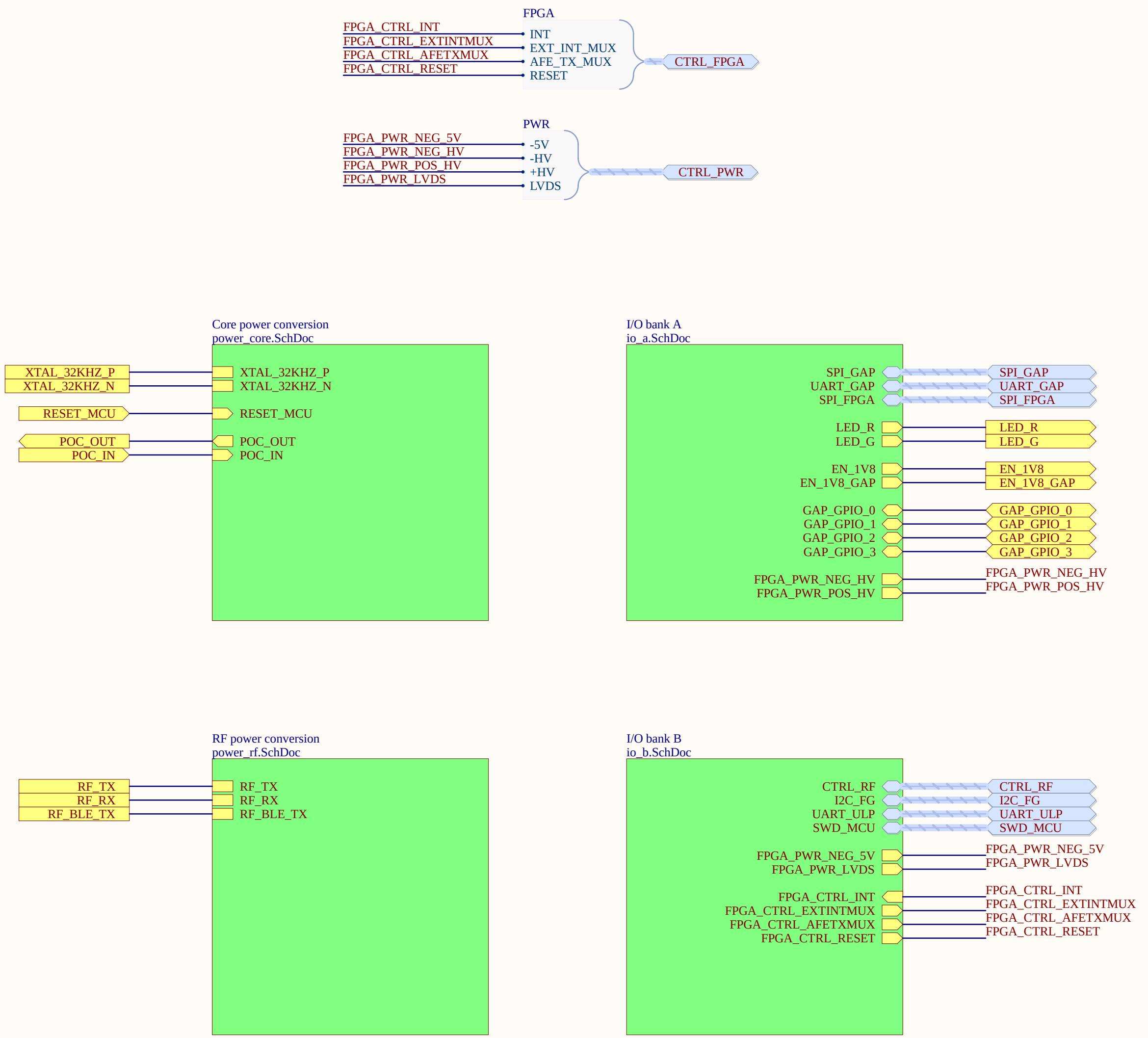
File:



1. Place all the decoupling capacitors close to the IC pins.
2. It is mandatory to follow the reference schematics for optimal RF performance.
3. Recommended to follow Star routing for the supply pins from Main Supply Source
4. IO_VDD_1, IO_VDD_2,IO_VDD_3, ULP_IO_VDD can be powered independently by different Voltage Sources based on their corresponding signals voltage levels requirements. Voltages must be as per the recommended Operating conditions"
5. If 917 IC variant with internal stacked flash is used, IO_VDD_1 supply can be either 1.8V or 3.3V based on the peripherals connected on GPIO 6:12 and GPIO 46:57.If external PSRAM or external Flash is connected on GPIO 46:57, then the signalling voltage on GPIO_6 to GPIO_12 must be same as external PSRAM/Flash supply voltage.
6. Use recommended MPNs(Manufacturer Part Number) shown near to components.
7. Even if GPIOs are not used, their respective IO domains must still be connected to the power supply

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Project: WiUS

Power & I/O

High-Throughput, Low-Power WiFi for Wearable Ultrasound Systems

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Date: 23.04.2024 17:23:44
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Variation: all_1v8

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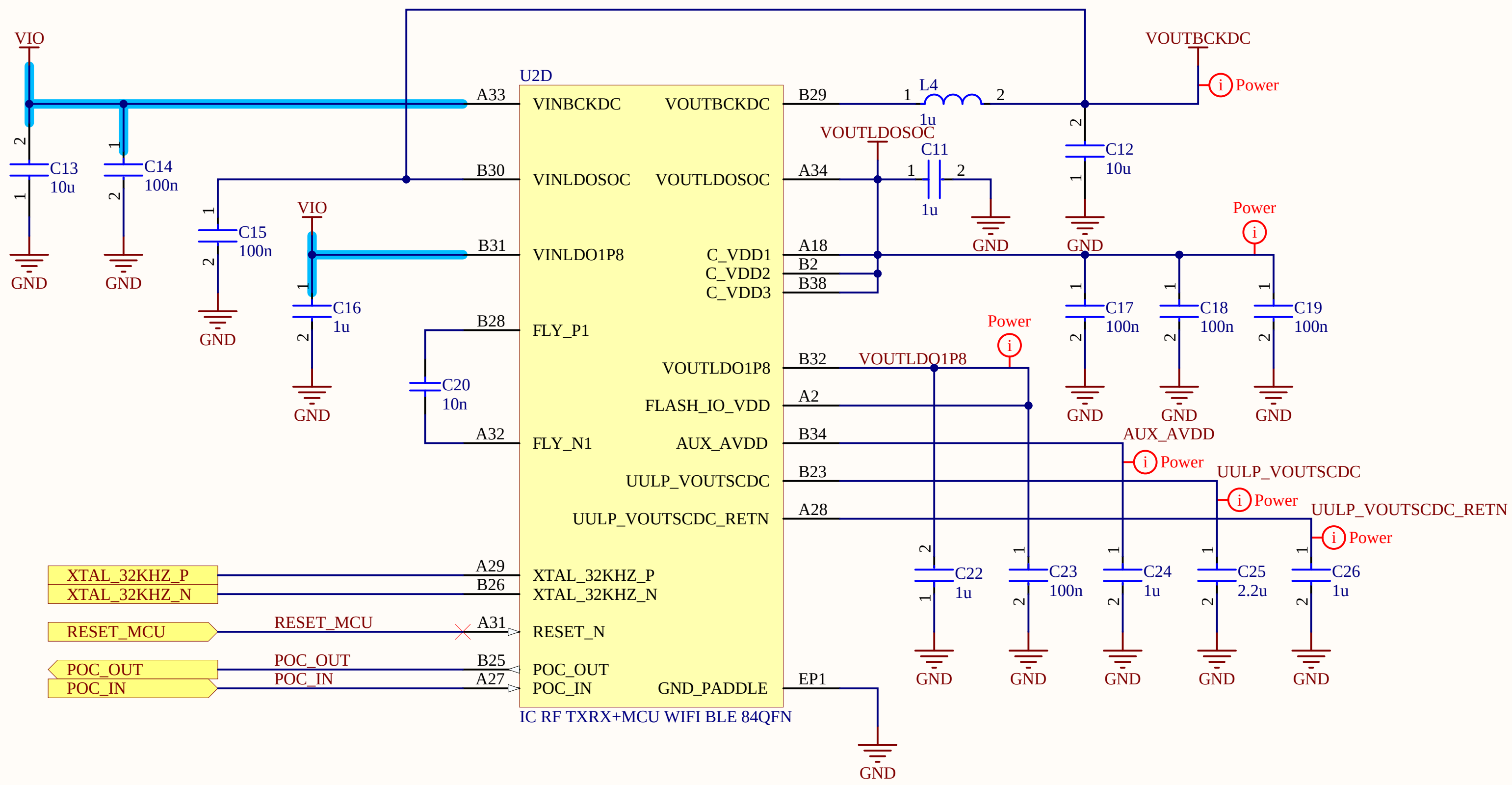

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Project: WiUS

Core power conversion

High-Throughput, Low-Power WiFi for Wearable Ultrasound Systems

Laboratory: Integrated Systems Laboratory

Date: 23.04.2024 17:23:44

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Variation: all_1v8

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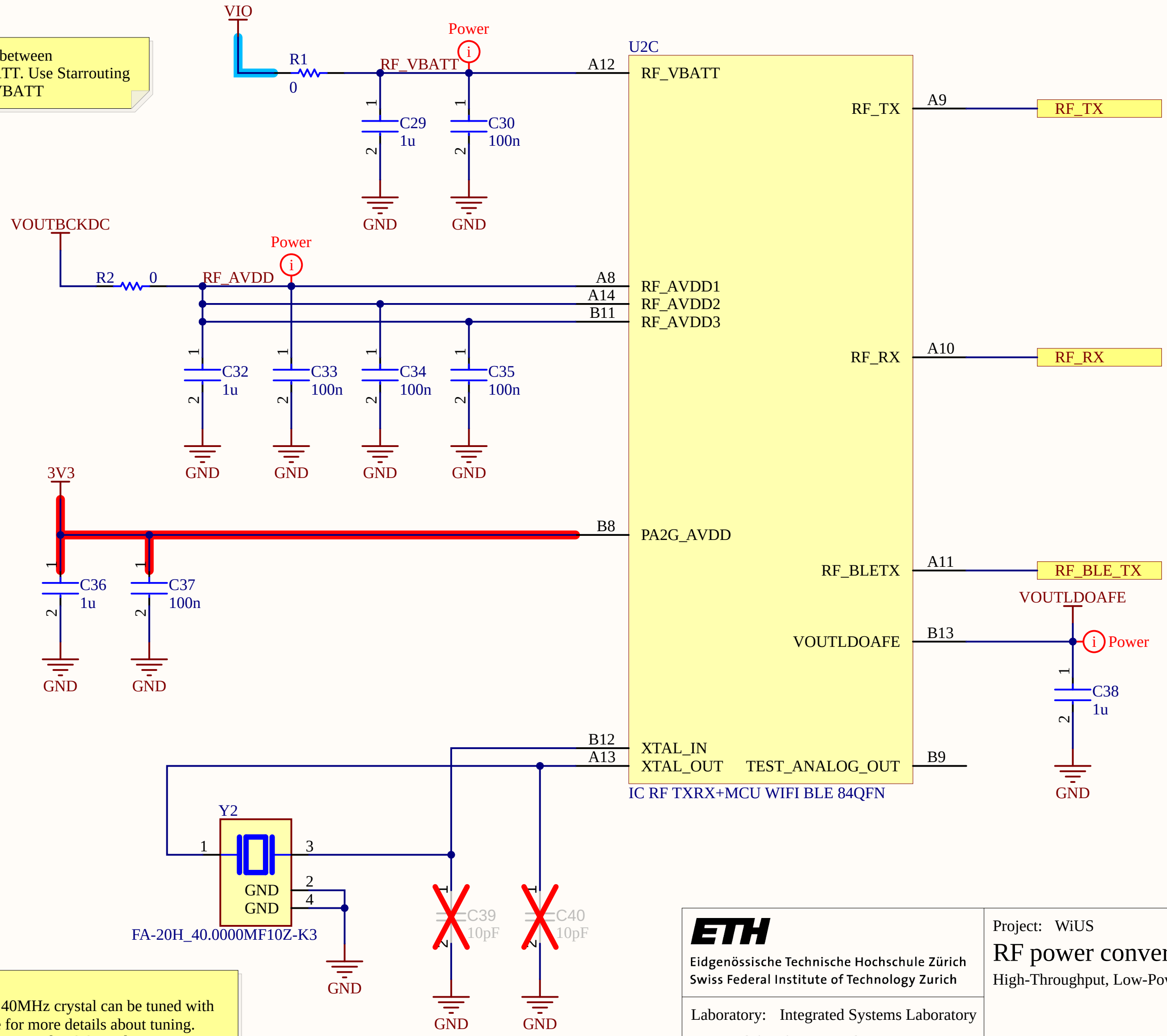
Do not share same Routing between PA2G_AVDD and RF_VBATT. Use Starrouting for PA2G_AVDD and RF_VBATT



1. VOUTBCKDC to be star routed to RF_AVDD Pins.
2. Use recommended Part for High Frequency Crystal (40MHz).This 40MHz crystal can be tuned with SoC internal capacitors. Refer to Crystal Calibration Application note for more details about tuning.
3. PA2G_AVDD can be independent of other supplies and it must be operated at 3.3V only.
4. C39 , C40 are optional tuning capacitors.
5. R1 , R2 are placeholder components for power supply noise filtering purpose.

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Project: WiUS

RF power conversion

High-Throughput, Low-Power WiFi for Wearable Ultrasound Systems

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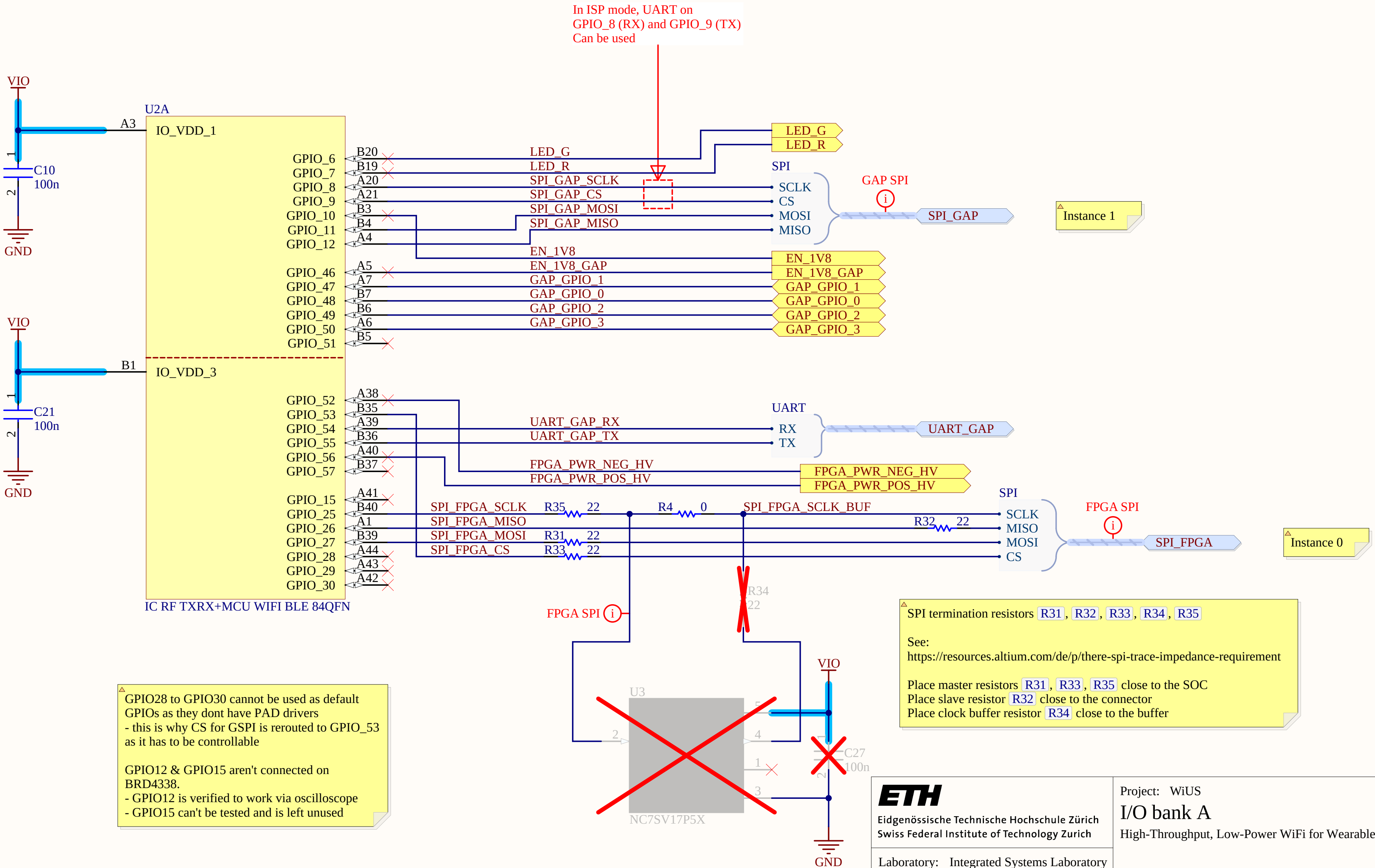

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Project: WiUS

I/O bank A

High-Throughput, Low-Power WiFi for Wearable Ultrasound Systems

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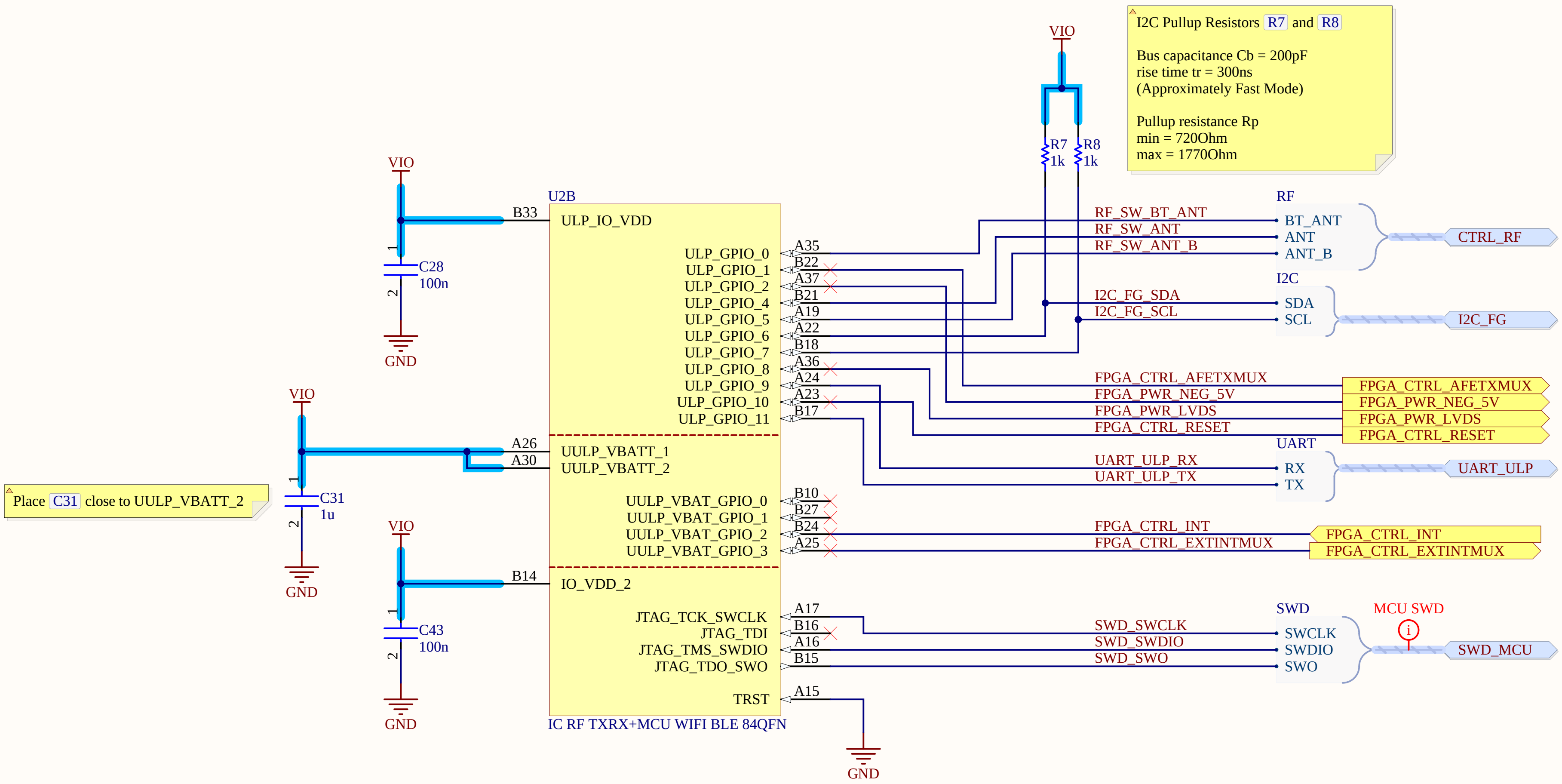

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Project: WiUS

I/O bank B

High-Throughput, Low-Power WiFi for Wearable Ultrasound Systems

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Variation: all_1v8

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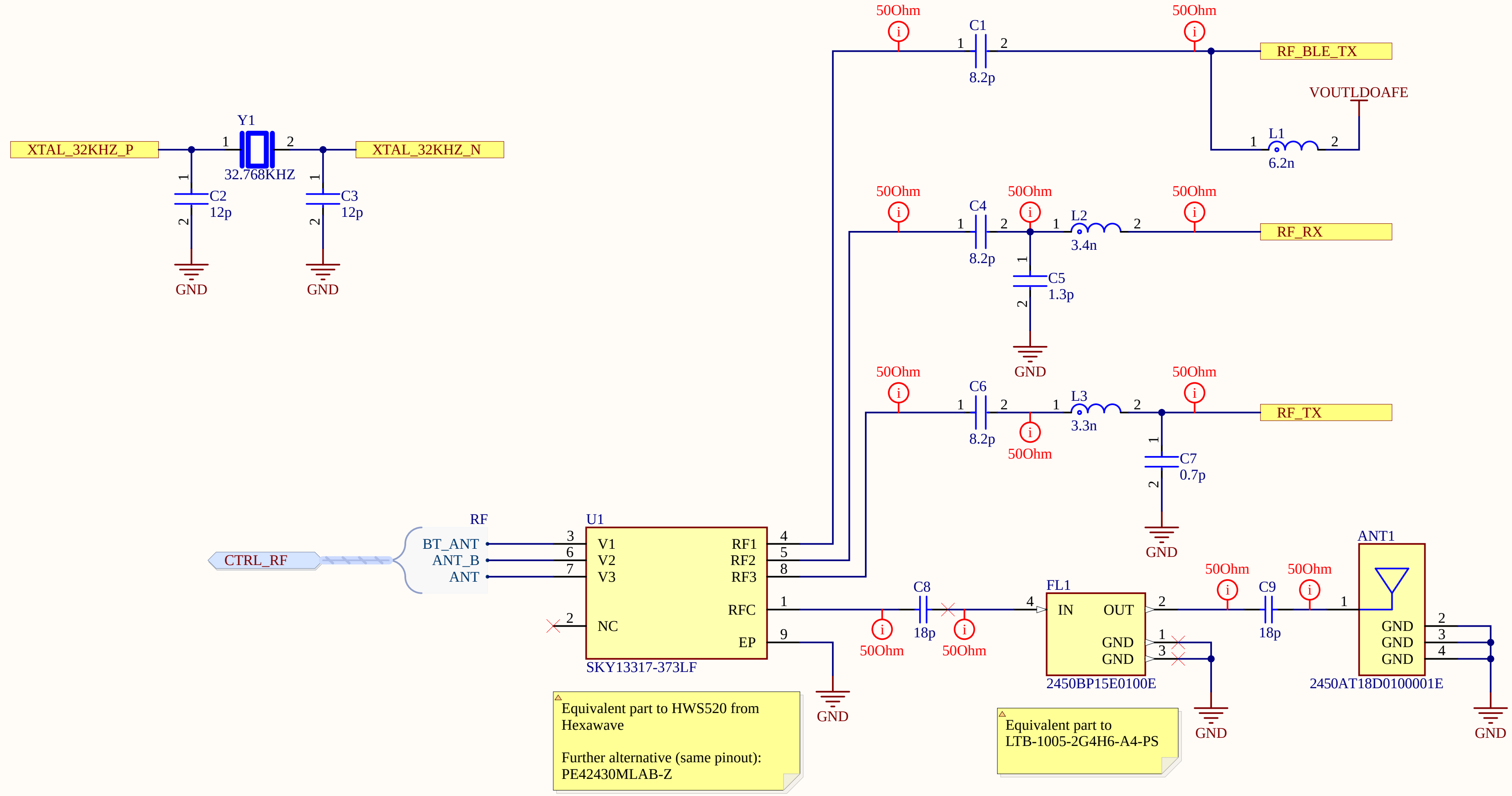

1. Maintain 50 ohm characteristic impedance for RF traces.
2. Additional matching circuit to be provided near the antenna, based on antenna used and location on the board.
3. For external RF switch option, follow the reference Layout for RF Front-end circuit from IC pins upto atleast RF Switch.
4. Follow the Gain offset calibration Application note for calibrating the power of RF front-end circuitry.
5. Follow the RF Matching and Layout Design Guide Application note for RF design related aspects.
6. There will be slight RF performance degradation when Antenna select signals (ULP_GPIO_4, ULP_GPIO_5, ULP_GPIO_0) are at 1.8V levels.

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Project: WiUS

RF & clocks

High-Throughput, Low-Power WiFi for Wearable Ultrasound Systems

Laboratory: Integrated Systems Laboratory

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Variation: all_1v8

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